

Day 1 — Amazon Redshift Basics & Hands-On

Topics Covered

- ✦ Data Warehouse Basics
- ✦ OLTP vs OLAP
- ✦ Introduction to Amazon Redshift
- ✦ Columnar Storage
- ✦ MPP Architecture
- ✦ Redshift Serverless Setup
- ✦ Query Editor v2
- ✦ Basic SQL Operations

Redshift Architecture

```
graph TD
    Applications --> Amazon_S3[Amazon S3]
    Amazon_S3 --> Spark_EMR[Spark / EMR]
    Spark_EMR --> Amazon_Redshift[Amazon Redshift]
    Amazon_Redshift --> BI_Dashboards[BI Dashboards]
```

Step 1 — Create Serverless Workgroup

Workgroup Details

Setting	Value
Workgroup Name	redshift-workgroup
Base Capacity	128 RPUs
Track	Current
IP Type	IPv4
SSL	Enabled

Setting	Value
Enhanced VPC Routing	Enabled

Network Configuration

Component	Value
VPC	vpc-06b7b0401fe6dbe7e
Security Group	sg-091014a6a26dbc631

Selected Subnets

```
subnet-08001f86dc890cec7
subnet-0b41bf8f2556bfd1
subnet-06324a4cc953fa24e
```

Step 2 — Create Namespace

Property	Value
Namespace	dev
Database	dev
Username	admin
Password	Manual Setup

✓ Namespace created successfully

Step 3 — Connect Query Editor v2

Steps Performed

- ✓ Open **Query Editor v2**
- ✓ Select the created workgroup
- ✓ Enter username & password that provided at the time of namespace creation
- ✓ Connected successfully

SQL Hands-On

Employees Table

```
CREATE TABLE employees (  
    emp_id INT,  
    name VARCHAR(50),  
    salary INT  
);
```

+ Insert Records

```
INSERT INTO employees  
VALUES  
(1, 'Rohan', 50000),  
(2, 'Amit', 70000),  
(3, 'Neha', 90000);
```

Fetch Data

```
SELECT * FROM employees;
```

Key Concepts Learned

Columnar Storage

- Stores data column-wise
- Faster analytical queries
- Better compression

MPP (Massively Parallel Processing)

- Queries run in parallel
- Faster execution on huge datasets

OLTP vs OLAP

OLTP	OLAP
MySQL	Redshift
Transactions	Analytics
Row-Based	Column-Based

Day 1 Outcome

 Successfully:

- Created Redshift Serverless setup
- Connected Query Editor v2
- Executed SQL queries
- Learned Redshift fundamentals

Screenshots

Hands-on screenshots added in repository.
